BKT and liquid–gas transitions from S_{QQ}

Progress and limitations, 28 October 2026. 2D two-component log plasma, Monte Carlo.

Where we are

Done

- **S_{QQ} pipeline at N = 50.** The 9.21 scan covers 10 densities, T* from 0.15 to 0.35 and E0* from 0 to 0.5, with 121 runs per density.
- **Wide pilot at ρ* = 0.05.** T* runs from 0.05 to 0.55 and E0* from 0 to 1.0, with hot and cold starts on a coarse 6 × 5 grid.
- **Two transition lines, both defined from S_{QQ} alone.** The BKT line is where 1/ε crosses 4T*. The liquid–gas line is the E0* step where $1/\varepsilon(k_{min})$ rises fastest at fixed T*. At ρ* = 0.05 the second line lands on the same step as the steepest S_{NN} jump.
- **Interactive page,** published as a new artifact: box animation, state table and cut charts.
- **Cross-checks.** The pilot $S_{QQ}(k)$ was compared with the reference figure. The potential was confirmed from the code: log-Coulomb, plus the cosh attraction, plus a hard core, with K = 1 in code units.
- **New data received.** N = 100 and N = 200 are in Downloads/data_n100_200. They are surveyed and complete, but not yet processed.

Data in hand

Set	N	Density ρ*	E0*	T*	Runs
9.21 scan	50	10 values, 0.001 to 0.2	0 to 0.5, step 0.05	0.15 to 0.35, step 0.02	121 per density
10.27 pilot	50	0.05	0 to 1.0, 5 values	0.05 to 0.55, 6 values	60 (hot and cold)
New: original	100 and 200	10 values, one batch each	0 to 0.5, step 0.05	0.15 to 0.35, step 0.02	121 per batch
New: bkt	100 and 200	0.001 only	0 to 0.15, step 0.05	0.15 to 0.35, step 0.01	84 each

Disk size: N = 50 is 1.5 GB, N = 100 is 2.1 GB and N = 200 is 4.0 GB unpacked (0.8 GB and 1.2 GB as tar.gz).

Limitations

Data

- **Nothing below T* = 0.15 is usable.** The pilot's T* = 0.05 row is unequilibrated. This is where the BKT and liquid–gas lines should meet, so the junction is unresolved.
- **The new N = 100/200 data does not fix that.** It has the same T* range as before, so it only refines lines we already have.
- **The grid is coarse.** The steps are 0.05 in E0* and 0.02 in T*, so each transition is a bracket, not a located point. The pilot is coarser still.

BKT line

- **Most drawn points are not real BKT points.** At $\rho^* \geq 0.03$ there are no filled (homogeneous) points. Beyond $E0^* \approx 0.30$ every crossing is inside a droplet, except at $\rho^* = 0.001$.
- **The current page still draws them.** It has not been changed yet. The fix is to cap the line at $E0^* \approx 0.30$ and draw only the filled points.

Liquid–gas line

- **It is a location of fastest change, not proof of a first-order transition.** Confirming one needs bimodality or hysteresis, or the S_{NN} Porod tail.
- **The finite box gives a rise, not the bulk drop.** The condensed phase is a neutral droplet, so $1/\varepsilon(k_{min})$ rises. The bulk liquid in the reference figure drops instead.
- **Overscreening is not reproduced.** The dip of $1/\varepsilon$ below 0 near $ka \approx 3.4$ did not appear. The minimum is about 0.6 to 0.7.
- **Low density is unreliable.** At $\rho^* \leq 0.01$ the lines are scattered.

Method

- **The shape classifier's cutoff is soft.** The 0.4 cutoff between crossover and condensed is not sharp.
- **Hot and cold starts disagree in places.** In the pilot, 6 of 30 cells differ by more than 0.15 in $1/\varepsilon$.

Physics still open

- **The boundary condition is unchecked.** The Ewald dipole term fixes it. It has not been compared with Elio's group (vacuum versus tin-foil).
- **K = 1 is inferred from how the code uses T*,** not written anywhere in the code. The BKT point at $T^* \approx 0.24$ supports it.

Pending notebook fixes (9.21_Density Correlation)

- VS Code needs "Revert File" for S_NN_Liquid_Gas.ipynb and Dielectric_Phase_Diagram.ipynb.
- The Porod text in section 4 is wrong.
- Step A, S(k) out to $k\sigma \approx 10$, is waiting for approval.
- The colour scale should be centred at 4, not 3.

Next

The most valuable step is the low-T* scan. The choices still needed are the densities, the T* and E0* ranges, the run length, and hot and cold starts. Processing N = 100 and 200 is optional.